



Terminology Translation

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Motivation

PROBLEM

- Enterprise Machine Translation (MT) must preserve domain-specific terminology (UI labels, product names, and technical concepts) because generic MT can mistranslate it, leading to inconsistent product documentation.

SETUP

- Independent TUM course project based on the **WMT25 Terminology Shared Task** sentence-level track.
- Translation is guided by explicit **terminology constraints** (source-term → target-term pairs) provided with each sentence.

RESEARCH QUESTIONS

- Do explicit **terminology constraints** in the prompt improve BLEU, chrF, term accuracy and consistency?
- Does *expanding* the terminology-constraint list (GPT-suggested vs. external dictionary) improve performance, and does cleaning that list help further?
- Does LoRA fine-tuning close the gap between open Qwen models and GPT-4o-mini?

Data and Task

JSONL Task Format

```
{
  "en": "Open the consumption model...",
  "de": "Öffnen Sie das Verbrauchsmodell...",
  "proper_terms": "consumption model -> Verbrauchsmodell,
  "random_terms": "Open -> Öffnen
}
```

Field legend

- en - English source
- de/rules - reference
- proper_terms - domain IT pairs
- random_terms - control (non-domain)

Data & Task

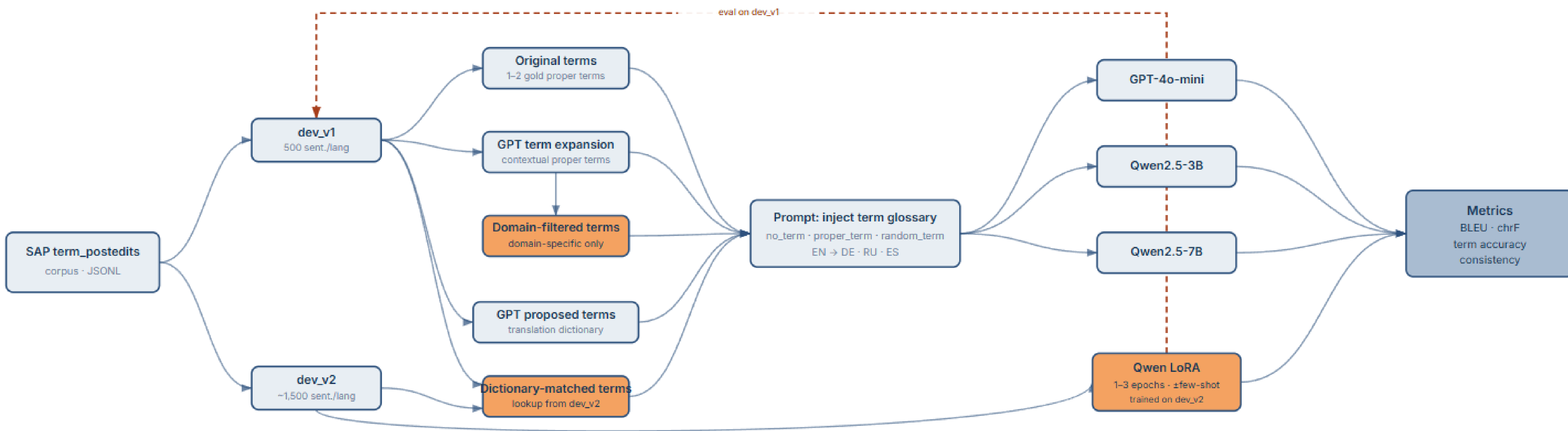
- dev_v1**: 500 sentences per language (test set).
- dev_v2**: ~1,500 SAP post-edited sentences per target language from SAP **term_postedits**; not the original shared-task dataset, but the closest available proxy to dev_v1 because both come from the same SAP documentation source.
- JSONL contains: English source, reference translation, **proper_terms**, and **random_terms** as a control.
- Language pairs: EN→DE, EN→RU, EN→ES.

Metrics

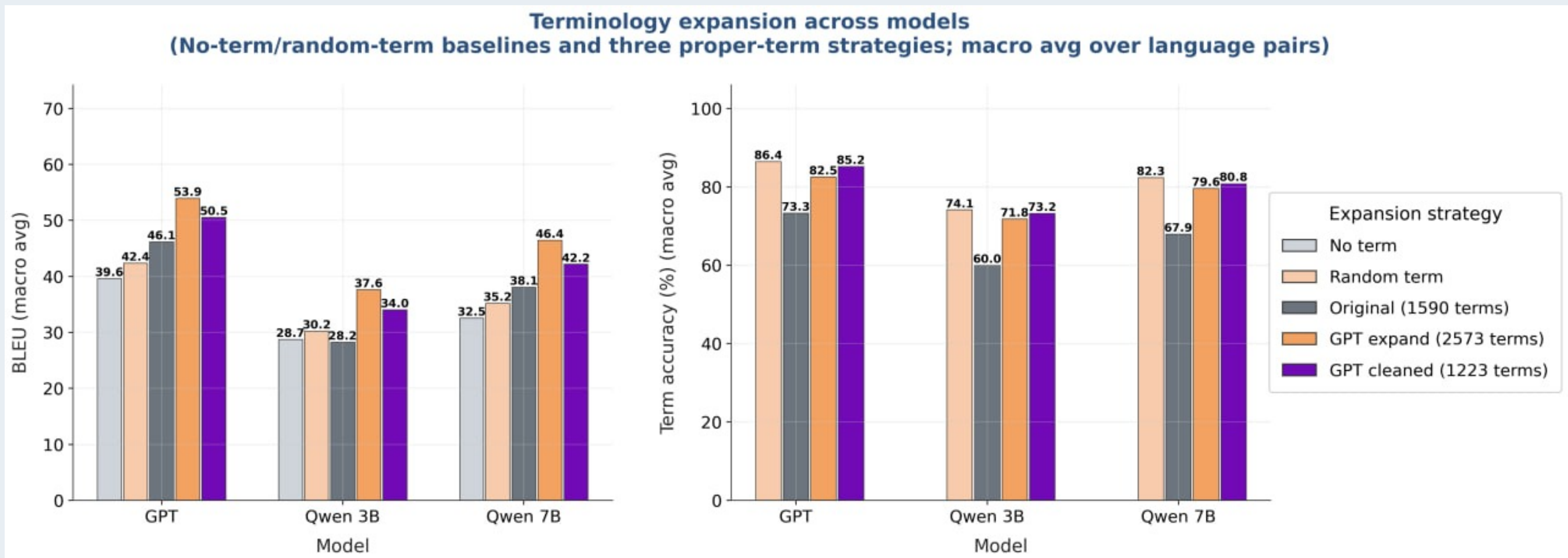
- BLEU / chrF** – general translation similarity; *insensitive to terminology preservation*.
- Term accuracy %** – percentage of provided proper terms correctly rendered in the output.
- Macro consistency** – mean per-term translation consistency.
- Weighted consistency** – frequency-weighted consistency.

Methods

Methods overview — shared pipeline, three study paths

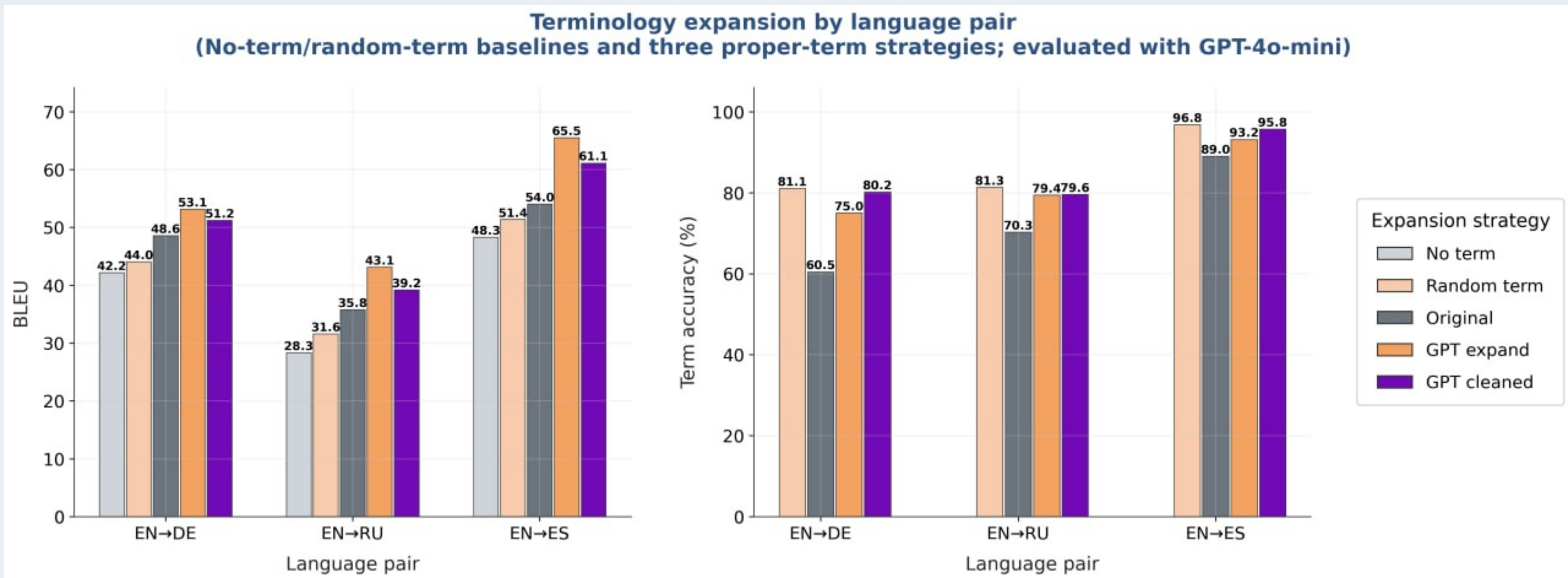


Term expansion strategies by model



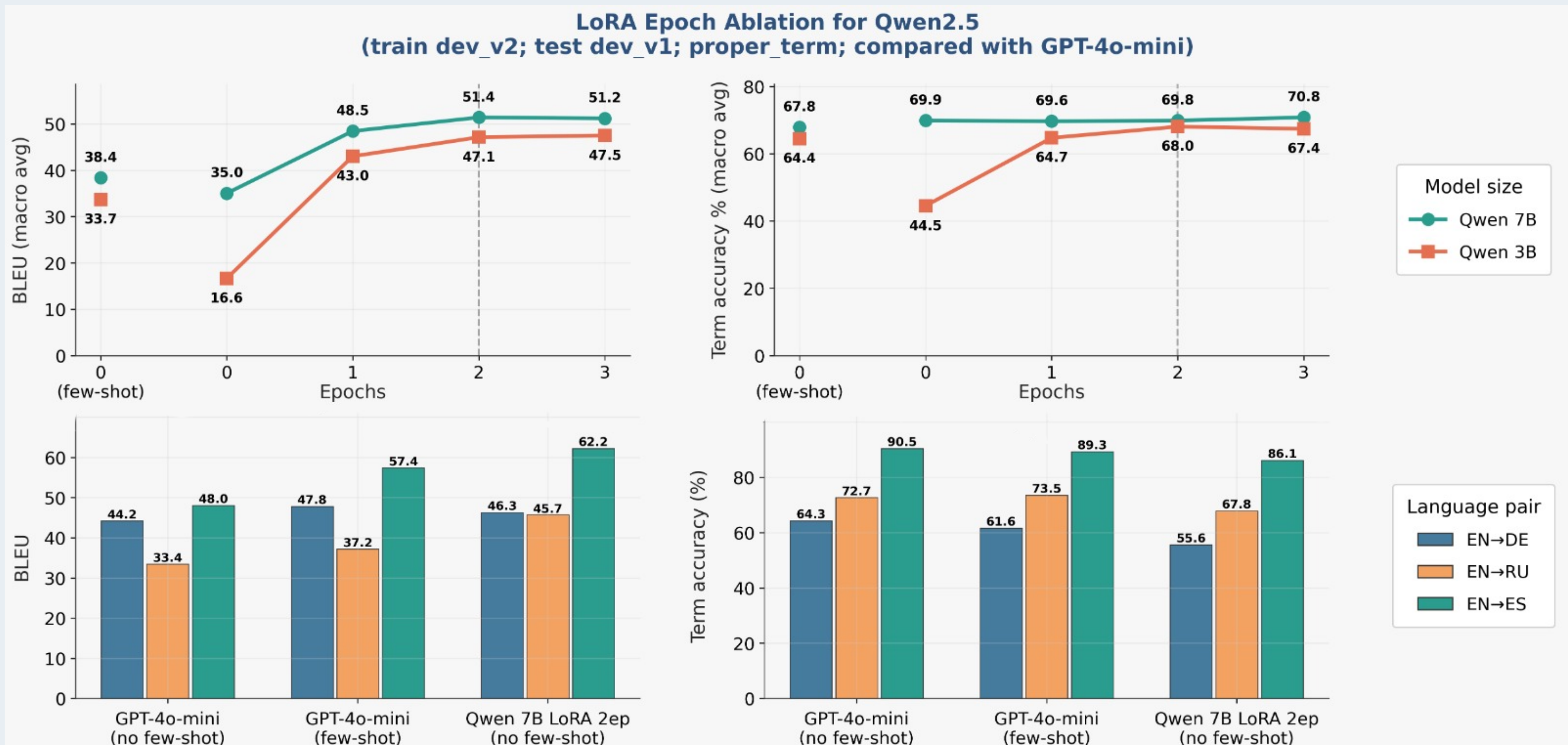
- Evaluated with **GPT-4o-mini** and base Qwen models with **3-shot prompting** per language.
- A **no-term baseline** on the BLEU panel shows that injecting proper terms improves translation quality for all models.
- GPT expand** (larger term list) leads on BLEU, while **GPT cleaned** (smaller, domain-filtered list) yields the highest term accuracy.

Term expansion strategies by language pair



- On BLEU, proper-term injection clearly beats the **no-term baseline** in every pair.
- EN→ES is strongest on both metrics.
- EN→DE outperforms EN→RU on BLEU, while EN→RU leads on term accuracy.
- Language difficulty** affects general quality; expansion choice shapes **terminology fidelity**.

Fine-Tuning: Best Models



- LoRA clearly improves **BLEU** over the Qwen base; gains build through epochs and peak around **2 epochs** (dashed line).
- Term accuracy stays relatively stable for **7B**; **3B** drops without few-shot prompting but recovers after fine-tuning – suggesting **few-shot prompting matters for terminology about as much as LoRA**.
- GPT-4o-mini vs. best Qwen 7B LoRA 2ep**: LoRA closes much of the BLEU gap, especially on EN→RU, while GPT few-shot remains strongest on **term accuracy**.
- Caveat**: some dev_v1 test sentences overlap lexically with dev_v2 training data.

Qualitative Examples

“It might make sense to hibernate this *space* as it’s not used much.”
→ “Es ist ggf. sinnvoll, diesen *Space* in den Ruhezustand zu setzen, da er nur wenig verwendet wird.”
space → Platz, Space

“Create new *category*.”
→ “Neuen *Typ* anlegen.”
category → Kategorie, Typ

“Spaces send *queries* or *jobs* to the database.”
→ “Spaces senden *Abfragen* oder *Jobs* an die Datenbank.”
query → Anfrage, Abfrage
job → Arbeit, Job

References

- Semenov et al. (2025). Findings of the WMT25 Terminology Translation Task: Terminology is Useful Especially for Good MTs. In *Proceedings of the Tenth Conference on Machine Translation*, Suzhou, China. Association for Computational Linguistics.
- SAP SE. *Software Documentation Data Set for Machine Translation: term_postedits/test_data*. GitHub repository, accessed 6 July 2026. https://github.com/SAP/software-documentation-data-set-for-machine-translation/tree/master/term_postedits/test_data